Weitzenböck formulas on Poisson probability spaces

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Abstract

This paper surveys and compares some recent approaches to stochastic infinite-dimensional geometry on the space Γ of configurations (i.e. locally finite subsets) of a Riemannian manifold M under Poisson measures. In particular, different approaches to Bochner-Weitzenböck formulas are considered. A unitary transform is also introduced by mapping functions of n configuration points to their multiple stochastic integral.

Key words: Poisson probability measure, Bochner-Weitzenböck formulas, stochastic analysis.
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1 Weitzenböck formula under a measure

Let M be a Riemannian manifold with volume measure dx, covariant derivative ∇, and exterior derivative d. Let ∇_μ^* and d_μ^* denote the adjoints of ∇ and d under a measure μ on M of the form $\mu(dx) = \mathrm{e}^{\phi(x)}dx$. The classical Weitzenböck formula under the measure μ states that

$$\mathrm{d}_\mu^*\mathrm{d} + \mathrm{d}\mathrm{d}_\mu^* = \nabla_\mu^*\nabla + R - \mathrm{Hess}\ \phi,$$

where R denotes the Ricci tensor on M. In terms of the de Rham Laplacian $H_R = \mathrm{d}_\mu^*\mathrm{d} + \mathrm{d}\mathrm{d}_\mu^*$ and of the Bochner Laplacian $H_B = \mathrm{d}_\mu^*\mathrm{d} + \mathrm{d}\mathrm{d}_\mu^*$ we have

$$H_R = H_B + R - \mathrm{Hess}\ \phi.$$

In particular the term Hess ϕ plays the role of a curvature under the measure μ.

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2 Probability: Poisson space

In this section we recall some facts on random functionals on Poisson space. The Poisson probability measure on $\mathbb{N}$ can be introduced by considering N independent $\{0,1\}$-valued Bernoulli random variables $X_1, \ldots, X_N$, with parameter λ/N, $\lambda > 0$. Then $X_1 + \cdots + X_N$ has a binomial law, and

$$P(X_1 + \cdots + X_N = k) = \binom{N}{k} \left(\frac{\lambda}{N}\right)^k \left(1 - \frac{\lambda}{N}\right)^{N-k}$$

converges to $\frac{\lambda^k}{k!}\mathrm{e}^{-\lambda}$ as N goes to infinity. This defines a probability measure π_λ on $\mathbb{N}$ as

$$\pi_\lambda(\{k\}) = \frac{\lambda^k}{k!}\mathrm{e}^{-\lambda}.$$

Let X be a metric space with a σ-finite Borel measure σ. The measure π_λ has the convolution property $\pi_\lambda \star \pi_\mu = \pi_{\lambda+\mu}$, which allows to construct the Poisson measure π_σ with intensity σ on

$$\Gamma = \left\{ \gamma = \sum_{k=1}^{k=n} \delta_{x_k}; \quad x_1, \ldots, x_n \in X, \ n \in \mathbb{N} \cup \{\infty\} \right\}$$

by letting

$$\pi_\sigma(\{\gamma \in \Gamma; \quad \gamma(A_1) = k_1, \ldots, \gamma(A_n) = k_n\}) = \frac{\sigma(A_1)^{k_1}}{k_1!}\mathrm{e}^{-\sigma(A_1)} \cdots \frac{\sigma(A_n)^{k_n}}{k_n!}\mathrm{e}^{-\sigma(A_n)},$$

where $A_1, \ldots, A_n$ are disjoint compact subsets of X. This measure is characterized by its Fourier transform

$$\int_\Gamma \mathrm{e}^{i\int_X f(x)d\gamma(x)} d\pi(\gamma) = \exp\left(\int_X (\mathrm{e}^{if(x)} - 1) d\sigma(x)\right).$$

If $\gamma \in \Gamma$ is finite with cardinal $|\gamma| = n$ we write

$$\gamma = \sum_{i=1}^{i=n} \delta_{x_i}.$$

Given Λ a compact subset we consider $F : \Gamma \to \mathbb{R}$ such that $F(\gamma) = F(\gamma \cap \Lambda)$, and written as

$$F(\gamma) = F(\gamma \cap \Lambda) = \mathrm{e}^{\sigma(\Lambda)/2} \sum_{n=0}^{\infty} 1_{\{|\gamma \cap \Lambda| = n\}} n! f_n(x_1, \ldots, x_n) = \sum_{n=0}^{\infty} J_n(f_n)$$

where f_n is a symmetric function with support in Λ^n, with

$$J_n(f_n)(\gamma) = J_n(f_n)(\gamma \cap \Lambda) = n! 1_{\{|\gamma \cap \Lambda| = n\}} \mathrm{e}^{\sigma(\Lambda)/2} f_n(x_1, \ldots, x_n), \quad n \geq 1.$$

The multiple Poisson stochastic integral of f_n is defined as

$$I_n(f_n) = \int_{\{(x_1,\ldots,x_n) \in X^n; \quad x_i \neq x_j, \ i \neq j\}.} f_n(x_1, \ldots, x_n)(\gamma - \sigma)(dx_1) \cdots (\gamma - \sigma)(dx_n),$$

and extends to $f_n \in L^2_\sigma(X)^{\circ n}$ via the well-known isometry

$$\int_\Gamma I_n(f_n) I_m(g_m) d\pi = n! 1_{\{n=m\}} \langle f_n, g_m \rangle_{L^2_\sigma(X)^{\circ n}}, \quad f_n \in L^2_\sigma(X)^{\circ n}, \ g_m \in L^2_\sigma(X)^{\circ m}.$$

We introduce a combinatorial transform $\tilde{K}$ which has some similarities with the K-transform, cf. [6] and references therein. The transform $\tilde{K}$ identifies the functional $J_n(f_n)$, which makes sense only in finite volume, to $I_n(f_n)$ which is defined for all square-integrable f_n.

Proposition 2.1 *The operator $\tilde{K}$ defined by*

$$\tilde{K} J_n(f_n) = I_n(f_n), \quad f_n \text{ symmetric in } \mathcal{C}_c(\Lambda^n), \ n \in \mathbb{N},$$

is unitary on $L^2_{\pi_\sigma}(\Gamma)$. Moreover, $\tilde{K}$ satisfies

$$\tilde{K} F(\gamma) = \sum_{\eta \subset \gamma} \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} \int_{X^k} F(\eta \cup \{y_1, \ldots, y_k\}) \sigma(dy_1) \cdots \sigma(dy_k).$$

Proof. We have

$$\begin{aligned}
\int_\Gamma J_n(f_n) J_m(g_m) d\pi &= n!^2 1_{\{n=m\}} \mathrm{e}^{\sigma(\Lambda)} \int_\Gamma 1_{\{|\gamma \cap \Lambda| = n\}} f_n(x_1, \ldots, x_n) g_n(x_1, \ldots, x_n) d\pi(\gamma) \\
&= \frac{n!^2}{\sigma(\Lambda)^n} \mathrm{e}^{\sigma(\Lambda)} \pi_\sigma(|\gamma \cap \Lambda| = n) \langle f_n, g_n \rangle_{L^2_\sigma(X)^{\otimes n}} \\
&= n! 1_{\{n=m\}} \langle f_n, g_n \rangle_{L^2_\sigma(X)^{\otimes n}},
\end{aligned}$$

which shows the first statement. On the other hand we have

$$\begin{aligned}
&\tilde{K} J_n(f_n)(\gamma) \\
&= \sum_{\eta \subset \gamma \cap \Lambda} \sum_{k=0}^{\infty} (-1)^k \frac{n!}{k!} \int_X 1_{\{|\eta \cup \{y_1,\ldots,y_k\}| = n\}} f_n(\eta \cup \{y_1, \ldots, y_k\}) \sigma(dy_1) \cdots \sigma(dy_k)
\end{aligned}$$

$$
\begin{aligned}
&= \sum_{k=0}^{k=n}(-1)^k\frac{n!}{k!}\sum_{\substack{\eta\subset\gamma\cap\Lambda\\|\eta|=n-k}}\int_X f_n(\eta\cup\{y_1,\dots,y_k\})\sigma(dy_1)\cdots\sigma(dy_k)\\
&= \sum_{k=0}^{k=n}(-1)^k\binom{n}{k}\sum_{\substack{x_1,\dots,x_{n-k}\in\gamma\cap\Lambda\\x_i\neq x_j,\ i\neq j}}\int_X f_n(\{x_1,\dots,x_{n-k},y_1,\dots,y_k\})\sigma(dy_1)\cdots\sigma(dy_k)\\
&= I_n(f_n)(\gamma),
\end{aligned}
$$

the last relation follows from e.g. Prop. 4.1 of [9]. □

If Λ is compact and $F(\gamma)=F(\gamma\cap\Lambda)$ we have

$$
\int_\Gamma F(\gamma)d\pi(\gamma)=\mathrm{e}^{-\sigma(\Lambda)}\sum_{n=0}^{\infty}\frac{1}{n!}\int_X\cdots\int_X f_n(x_1,\dots,x_n)\sigma(dx_1)\cdots\sigma(dx_n).
$$

In the particular case $X=\mathbb{R}_+$ with σ the Lebesgue measure, the standard Poisson process is defined as

$$
N_t(\gamma)=\gamma([0,t])=\sum_{k=1}^{\infty}1_{[T_k,\infty[}(t),\quad t>0,
$$

i.e. every configuration $\gamma\in\Gamma$ can be viewed as the ordered sequence $\gamma=(T_k)_{k\geq 1}$ of jump times of $(N_t)_{t\in\mathbb{R}_+}$ on $\mathbb{R}_+$. Let $f_n\in\mathcal{C}_c([0,\lambda]^n)$ be symmetric. Then

$$
\begin{aligned}
\int_\Gamma f_n(T_1,\dots,T_n)d\pi(\gamma)&=\mathrm{e}^{-\lambda}\sum_{k=n}^{\infty}\frac{1}{k!}\int_0^\lambda\cdots\int_0^\lambda f_n(t_1,\dots,t_n)\sigma(dt_1)\cdots\sigma(dt_k).\\
&= \sum_{k=n}^{\infty}\mathrm{e}^{-\lambda}\int_0^\lambda\int_0^{t_k}\cdots\int_0^{t_1} f_n(t_1,\dots,t_n)\sigma(dt_1)\cdots\sigma(dt_k).\\
&= \sum_{k=n}^{\infty}\mathrm{e}^{-\lambda}\int_0^\lambda\frac{(\lambda-t_n)^{k-n}}{(k-n)!}\int_0^{t_n}\cdots\int_0^{t_1} f_n(t_1,\dots,t_n)\sigma(dt_1)\cdots\sigma(dt_n).\\
&= \int_0^\infty\mathrm{e}^{-t_n}\int_0^{t_n}\cdots\int_0^{t_2} f_n(t_1,\dots,t_n)dt_1\cdots dt_n.
\end{aligned}
$$

This formula extends to f bounded and measurable.

3 Geometry

We recall the construction of [1], [2] in the case of 1-forms, see also [3] for the case of n-forms. We assume that X is a Riemannian manifold. The tangent space at $\gamma\in\Gamma$

is taken to be

$$L^2(X; TX, \gamma) = \oplus_{x\in\gamma} T_x X.$$

A differential form of order n maps $\gamma \in \Gamma$ into the antisymmetric tensor product

$$\wedge^n(T_\gamma \Gamma) = \wedge^n(\oplus_{x\in\gamma} T_x X).$$

Bochner and de Rham Laplacians on differential forms over configuration spaces are then constructed from their counterparts at the level of the manifold X. Let d_x^X be the exterior differential on X, let ∇_x^X, Δ_x^X be the natural covariant derivative and Bochner Laplacian on the bundle $T_{\gamma\setminus\{x\}\cup\{y\}}\Gamma \to y \in \mathcal{O}_{\gamma,x}$, where $\mathcal{O}_{\gamma,x}$ is an open set in X such that $\bar{\mathcal{O}}_{\gamma,x} \cap (\gamma \setminus \{x\}) = \emptyset$. The covariant derivative of the smooth differential 1-form W is defined as

$$(\nabla_x W_x(\gamma, x))_{x\in\gamma} \in T_\gamma\Gamma \otimes T_\gamma\Gamma,$$

where $W_x(\gamma, y) = W((\gamma \setminus \{x\}) \cup \{y\})$, $x, y \in X$. The Bochner Laplacian H^B on Γ is defined as

$$H^B W(\gamma) = -\sum_{x\in\gamma} \Delta_x^X W_x(\gamma, x).$$

The exterior derivative d^Γ is defined as

$$d^\Gamma W = \sum_{x\in\gamma} \sum_{y\in\gamma} d_x^X W_x(\gamma, x)_y,$$

where $W_x(\gamma, x)_y$ is the component of $W_x(\gamma, x)$ of index $y \in \gamma$, with adjoint

$$d^{\Gamma *} W = \sum_{x\in\gamma} \sum_{y\in\gamma} d_x^{X*} W_x(\gamma, x)_{x,y},$$

where $W_x(\gamma, x)_{x,y}$ is the component of $W_x(\gamma, x)$ of index (x, y) and d_x^{X*} is the adjoint of d_x^X under the volume element σ on X. A Weitzenböck formula is stated in [1], [3] as

$$H^R = H^B + R, \tag{3.1}$$

where H^R is the de Rham Laplacian $H^R = d^\Gamma d^{\Gamma *} + d^{\Gamma *} d^\Gamma$ and the curvature term

$$R(\gamma) = \sum_{x\in\gamma} R(\gamma, x)$$

has the explicit expression

$$R(\gamma,x)(V(\gamma)_y) = 1_{\{x=y\}} \sum_{i,j=1}^{d} \mathrm{Ric}_{ij}(x) e_i \langle V(\gamma)_x, e_j \rangle_x,$$

where $(e_j)_{j=1}^{j=d}$ is an orthonormal basis of T_xX. Formula (3.1) can be viewed as the lifting to Γ of the Weitzenböck formula on X.

Note that in the above construction the curvature term in (3.1) is essentially due to the curvature of X, in particular it vanishes if $X = \mathbb{R}^d$ and no curvature term is induced from the Poisson measure itself.

In this paper we present a different geometry on the infinite-dimensional space Γ, in which the Ricci curvature tensor under the Poisson measure appears to be the identity operator when $X = \mathbb{R}_+$, see [8] when X is a more general Riemannian manifold. .

Lifting of differential structure

Let $\mathcal{S}$ denote the space of cylindrical functionals of the form

$$F(\gamma) = f(T_1, \ldots, T_n), \quad f \in \mathcal{C}_b^\infty(\mathbb{R}^n). \tag{3.2}$$

Let $\mathcal{U}$ denote the space of smooth processes of the form

$$u(\gamma,x) = \sum_{i=1}^{i=n} F_i(\gamma) h_i(x), \quad (\gamma,x) \in \Gamma \times \mathbb{R}_+, \quad h_i \in \mathcal{C}_c^\infty(\mathbb{R}_+), \quad F_i \in \mathcal{S}, \quad i = 1, \ldots, n. \tag{3.3}$$

The differential geometric objects to be introduced below have finite dimensional counterparts, and each of them has a stochastic interpretation. The following table describes the correspondence between geometry and probability.

	Geometry	Probability
γ	element of Γ	point measure on $\mathbb{R}_+$
$\mathcal{C}_c^\infty(\mathbb{R}_+)$	tangent vectors to Γ	test functions on $\mathbb{R}_+$
σ	Riemannian metric on Γ	Lebesgue measure
d	gradient on Γ	stochastic gradient
$\mathcal{U}$	vector fields on Γ	stochastic processes
$\mathrm{d}u$	exterior derivative of $u \in \mathcal{U}$	two-parameter process
$\{\cdot,\cdot\}$	bracket of vector fields on Γ	bracket on $\mathcal{U} \times \mathcal{U}$
Ω	curvature tensor on Γ	trilinear mapping on $\mathcal{U}$
d^*	divergence on Γ	stochastic integral operator

Divergence operator

The definition of the following gradient operator goes back to [4].

Definition 3.1 *Given $F \in \mathcal{S}$, $F = f(T_1, \ldots, T_d)$, let*

$$\mathrm{d}_t F(\gamma) = -\sum_{k=1}^{k=d} 1_{[0,T_k]}(t) \partial_k f(T_1, \ldots, T_d), \quad t \geq 0.$$

The following is a finite-dimensional integration by parts formula for d.

Lemma 3.1 *We have for $F = f(T_1, \ldots, T_d)$ and $h \in \mathcal{C}_c(\mathbb{R}_+)$:*

$$\int_\Gamma \langle \mathrm{d}F, h \rangle_{L^2(\mathbb{R}_+)} d\pi(\gamma) = \int_\Gamma F(\gamma) \left(\sum_{k=1}^{k=d} h(T_k) - \int_0^{T_d} h(t)dt \right) d\pi(\gamma).$$

Proof. All $\mathcal{C}^\infty$ functions on $\Delta_d = \{(t_1, \ldots, t_d); \quad 0 \leq t_1 < \cdots < t_d\}$ are extended by continuity to the closure of Δ_d. We have

$$\int_\Gamma \langle \mathrm{d}F(\gamma), h \rangle_{L^2(\mathbb{R}_+)} d\pi(\gamma) = -\sum_{k=1}^{k=d} \int_0^\infty \mathrm{e}^{-t_d} \int_0^{t_d} \cdots \int_0^{t_2} \int_0^{t_k} h(s) ds \partial_k f(t_1, \ldots, t_d) dt_1 \cdots dt_d$$

$$= \int_0^\infty \mathrm{e}^{-t_d} \int_0^{t_d} \cdots \int_0^{t_2} h(t_1) f(t_1, \ldots, t_d) dt_1 \cdots dt_d$$

$$- \int_0^\infty \mathrm{e}^{-t_d} \int_0^{t_d} \cdots \int_0^{t_3} \int_0^{t_2} h(s) ds f(t_2, t_2, \ldots, t_d) dt_2 \cdots dt_d$$

$$
\begin{aligned}
&+\sum_{k=2}^{k=d}\int_0^\infty \mathrm{e}^{-t_d}\int_0^{t_d}\cdots\int_0^{t_2} h(t_k)f(t_1,\ldots,t_d)dt_1\cdots dt_d\\
&-\int_0^\infty \mathrm{e}^{-t_d}\int_0^{t_d} h(s)ds\int_0^{t_d}\cdots\int_0^{t_2} f(t_1,\ldots,t_d)dt_1\cdots dt_d\\
&-\sum_{k=2}^{k=d-1}\int_0^\infty \mathrm{e}^{-t_d}\int_0^{t_d}\cdots\int_0^{t_{k+1}}\int_0^{t_{k-1}}\cdots\int_0^{t_2}\\
&\qquad\int_0^{t_{k+1}} h(s)dsf(t_1,.,t_{k-1},t_{k+1},t_{k+1},.,t_d)dt_1\cdots dt_{k+1}dt_{k-1}\cdots dt_d\\
&+\sum_{k=2}^{k=d}\int_0^\infty \mathrm{e}^{-t_d}\int_0^{t_d}\cdots\int_0^{t_k}\int_0^{t_{k-2}}\cdots\int_0^{t_2}\int_0^{t_k} h(s)dsf(t_1,.,t_{k-2},t_k,t_k,.,t_d)dt_1\cdots dt_d\\
&=\sum_{k=1}^{k=d}\int_0^\infty \mathrm{e}^{-t_d}\int_0^{t_d}\cdots\int_0^{t_2} h(t_k)f(t_1,\ldots,t_d)dt_1\cdots dt_d\\
&\quad-\int_0^\infty \mathrm{e}^{-t_d}\int_0^{t_d} h(s)ds\int_0^{t_d}\cdots\int_0^{t_2} f(t_1,\ldots,t_d)dt_1\cdots dt_d\\
&=\int_\Gamma F(\gamma)\left(\sum_{k=1}^{k=d} h(T_k)-\int_0^{T_d} h(t)dt\right)d\pi(\gamma).
\end{aligned}
$$

□

The following definition of the divergence coincides with the compensated Poisson stochastic integral with respect to $(N_t-t)_{t\in\mathbb{R}_+}$ on the adapted square-integrable processes.

Definition 3.2 *We define* d_π^* *on* $\mathcal{U}$ *by*

$$
\mathrm{d}_\pi^*(hG)=\int_0^\infty h(t)(\gamma(dt)-dt)-\langle h,\mathrm{d}G\rangle_{L^2(\mathbb{R}_+)},\qquad G\in\mathcal{S},\ h\in L^2(\mathbb{R}_+).
$$

Using this definition, an integration by parts formula can be obtained independently of the dimension.

Proposition 3.1 *The divergence operator* $\mathrm{d}_\pi^*: L^2(\Gamma\times\mathbb{R}_+)\longrightarrow L^2(\Gamma)$ *is the closable adjoint of* d*, i.e.*

$$
\int_\Gamma F\mathrm{d}_\pi^* u\ d\pi(\gamma)=\int_\Gamma\langle \mathrm{d}F,u\rangle_{L^2(\mathbb{R}_+)}d\pi(\gamma),\quad F\in\mathcal{S},\quad u\in\mathcal{U}. \tag{3.4}
$$

Proof. Given Lemma 3.1 it suffices to notice that if $k > d$,

$$\begin{aligned}
\int_\Gamma F(\gamma)h(T_k)d\pi(\gamma) &= \int_0^\infty \mathrm{e}^{-t_k} h(t_k) \int_0^{t_k} \cdots \int_0^{t_d} \cdots \int_0^{t_2} f(t_1, \ldots, t_d) dt_1 \cdots dt_k \\
&= \int_0^\infty \mathrm{e}^{-t_k} \int_0^{t_k} h(s)ds \int_0^{t_k} \cdots \int_0^{t_d} \cdots \int_0^{t_2} f(t_1, \ldots, t_d) dt_1 \cdots dt_k \\
&\quad - \int_0^\infty \mathrm{e}^{-t_{k-1}} \int_0^{t_{k-1}} h(s)ds \int_0^{t_{k-1}} \cdots \int_0^{t_d} \cdots \int_0^{t_2} f(t_1, \ldots, t_d) dt_1 \cdots dt_{k-1} \\
&= \int_\Gamma F(\gamma) \int_{x_{k-1}}^{T_k} h(t) dt d\pi(\gamma),
\end{aligned}$$

in other terms the discrete-time process

$$\left(\sum_{k=1}^{k=n} h(T_k) - \int_0^{T_k} h(t)dt \right)_{k \geq 1} = \left(\int_0^{T_k} h(t) d(N_t - t) \right)_{k \geq 1}$$

is a martingale. Hence Relation (3.4) also implies that for $F, G \in \mathcal{S}$,

$$\begin{aligned}
\int_\Gamma \langle \mathrm{d}F, hG \rangle_{L^2(\mathbb{R}_+)} d\pi(\gamma) &= \int_\Gamma \langle \mathrm{d}(FG), h \rangle_{L^2(\mathbb{R}_+)} - F \langle \mathrm{d}G, h \rangle_{L^2(\mathbb{R}_+)} d\pi(\gamma) \qquad (3.5) \\
&= \int_\Gamma F \left(G \int_0^\infty h(t)(\gamma(dt) - dt) - \langle h, \mathrm{d}G \rangle_{L^2(\mathbb{R}_+)} \right) d\pi(\gamma) \\
&= \int_\Gamma F \mathrm{d}_\pi^*(hG) d\pi(\gamma). \qquad (3.6)
\end{aligned}$$

□

Covariant derivative

Given $u \in \mathcal{U}$ we define the covariant derivative $\nabla_u v$ in the direction $u \in L^2(\mathbb{R}_+)$ of the vector field $v = \sum_{i=1}^{i=n} F_i h_i \in \mathcal{U}$ as

$$\nabla_u v(t) = \sum_{i=1}^{i=n} h_i(t) \mathrm{d}_u F_i - F_i \dot{h}_i(t) \int_0^t u(s)ds, \qquad t \in \mathbb{R}_+, \qquad (3.7)$$

where

$$\mathrm{d}_u F = \langle \mathrm{d}F, u \rangle_{L^2(\mathbb{R}_+)}, \qquad F \in \mathcal{S}.$$

We have

$$\nabla_{uF}(vG) = Fv\mathrm{d}_u G + FG\nabla_u v, \quad u, v \in \mathcal{C}_c^\infty(\mathbb{R}_+), \quad F, G \in \mathcal{S}. \qquad (3.8)$$

We also let

$$\nabla_s v(t) = \sum_{i=1}^{i=n} h_i(t) \mathrm{d}_s F_i - F_i \dot{h}_i(t) 1_{[0,t]}(s), \qquad s, t \in \mathbb{R}_+,$$

in order to write

$$\nabla_u v(t) = \int_0^\infty u(s) \nabla_s v(t) ds, \quad t \in \mathbb{R}_+, \quad u, v \in \mathcal{U}.$$

Lie-Poisson bracket

Definition 3.3 *The Lie bracket* $\{u, v\}$ *of* $u, v \in \mathcal{C}_c^\infty(\mathbb{R}_+)$*, is defined to be the unique element of* $\mathcal{C}_c^\infty(\mathbb{R}_+)$ *satisfying* $(\mathrm{d}_u \mathrm{d}_v - \mathrm{d}_v \mathrm{d}_u) F = \mathrm{d}_w F$*,* $F \in \mathcal{S}$*.*

The bracket $\{u, v\}$ is defined for $u, v \in \mathcal{U}$ with

$$\{Fu, Gv\}(x) = FG\{u, v\}(x) + v(x) F \mathrm{d}_u G - u(x) G \mathrm{d}_v F, \quad x \in \mathbb{R}_+,$$

$u, v \in \mathcal{C}_c^\infty(\mathbb{R}_+)$, $F, G \in \mathcal{S}$.

Vanishing of torsion

Proposition 3.2 *The Lie bracket* $\{u, v\}$ *of* $u, v \in \mathcal{U}$ *satisfies*

$$\{u, v\} = \nabla_u v - \nabla_v u,$$

i.e. the connection defined by ∇ *has a vanishing torsion.*

Proof. We have $F(\gamma) = T_n$. If $u, v \in \mathcal{C}_c^\infty(\mathbb{R}_+)$ we have

$$\begin{aligned}
(\mathrm{d}_u \mathrm{d}_v - \mathrm{d}_v \mathrm{d}_u) T_n &= -\mathrm{d}_u \int_0^{T_n} v(s) ds + \mathrm{d}_v \int_0^{T_n} u(s) ds \\
&= v(T_n) \int_0^{T_n} u(s) ds - u(T_n) \int_0^{T_n} v(s) ds \\
&= \int_0^{T_n} \left(\dot{v}(t) \int_0^t u(s) ds - \dot{u}(t) \int_0^t v(s) ds \right) dt \\
&= \mathrm{d}_{\nabla_u v - \nabla_v u} T_n.
\end{aligned}$$

Since d is a derivation, this shows that

$$\mathrm{d}_u \mathrm{d}_v - \mathrm{d}_v \mathrm{d}_u = \mathrm{d}_{\nabla_u v - \nabla_v u}, \quad u, v \in \mathcal{U}.$$

The extension to $u, v \in \mathcal{U}$ follows from (3.8). □

Vanishing of curvature

Proposition 3.3 *The curvature tensor Ω of ∇ vanishes on $\mathcal{U}$, i.e.*

$$\Omega(u,v)h := [\nabla_u, \nabla_v]h - \nabla_{\{u,v\}}h = 0, \quad u, v, h \in \mathcal{U},$$

and $\mathcal{U}$ is a Lie algebra under the bracket $\{\cdot,\cdot\}$.

Proof. We have, letting $\tilde{u}(t) = -\int_0^t u(s)ds$:

$$[\nabla_u, \nabla_v]h = \tilde{u}\dot{\nabla_v h} - \tilde{v}\dot{\nabla_u h} = \tilde{u}\,\dot{\tilde{v}\dot{h}} - \tilde{v}\,\dot{\tilde{u}\dot{h}} = -\tilde{u}v\dot{h} + \tilde{v}u\dot{h},$$

and

$$\nabla_{\{u,v\}}h = \nabla_{\tilde{u}\dot{v}-\tilde{v}\dot{u}}h = \widetilde{(\tilde{u}\dot{v} - \tilde{v}\dot{u})}\dot{h} = (u\tilde{v} - v\tilde{u})\dot{h},$$

hence $\Omega(u,v)h = 0$, $h, u, v \in \mathcal{C}_c^\infty(\mathbb{R}_+)$. The extension of the result to $\mathcal{U}$ follows again from (3.8). The Lie algebra property follows from the vanishing of Ω. □

Exterior derivative

The exterior derivative $\mathrm{d}u$ of a smooth vector field $u \in \mathcal{U}$ is defined from

$$\langle \mathrm{d}u, h_1 \wedge h_2 \rangle_{L^2(\mathbb{R}_+)\wedge L^2(\mathbb{R}_+)} = \langle \nabla_{h_1} u, h_2 \rangle_{L^2(\mathbb{R}_+)} - \langle \nabla_{h_2} u, h_1 \rangle_{L^2(\mathbb{R}_+)},$$

$h_1, h_2 \in \mathcal{U}$. We have

$$\|\mathrm{d}u\|^2_{L^2(\mathbb{R}_+)\wedge L^2(\mathbb{R}_+)} = 2\int_0^\infty \int_0^\infty (\mathrm{d}u(s,t))^2 ds dt, \tag{3.9}$$

where

$$\mathrm{d}u(s,t) = \frac{1}{2}(\nabla_s u(t) - \nabla_t u(s)), \quad s, t \in \mathbb{R}_+, \quad u \in \mathcal{U}.$$

Isometry formula

Lemma 3.2 *We have for $u \in \mathcal{U}$:*

$$\int_\Gamma (\mathrm{d}_\pi^* u)^2 d\pi(\gamma) = \int_\Gamma \|u\|^2_{L^2(\mathbb{R}_+)} d\pi(\gamma) + \int_\Gamma \int_0^\infty \int_0^\infty \nabla_s u(t) \nabla_t u(s) ds dt d\pi(\gamma). \tag{3.10}$$

Proof. (cf. [8], [7] and the proof of [5] for path spaces over Lie groups). Given $u = \sum_{i=1}^n h_i F_i \in \mathcal{U}$ we have

$$\begin{aligned}
\int_\Gamma \mathrm{d}_\pi^*(h_i F_i)\mathrm{d}_\pi^*(h_j F_j)d\pi(\gamma) &= \int_\Gamma F_i \mathrm{d}_{h_i}\mathrm{d}_\pi^*(h_j F_j)d\pi(\gamma) \\
&= \int_\Gamma F_i \mathrm{d}_{h_i}(F_j \mathrm{d}_\pi^*(h_j) - \mathrm{d}_{h_j}F_j)d\pi(\gamma) \\
&= \int_\Gamma (F_i F_j \mathrm{d}_{h_i}\mathrm{d}_\pi^* h_j + F_i \mathrm{d}_\pi^*(h_j)\mathrm{d}_{h_i}F_j - F_i \mathrm{d}_{h_i}\mathrm{d}_{h_j}F_j)d\pi(\gamma) \\
&= \int_\Gamma (F_i F_j \langle h_i, h_j\rangle_{L^2(\mathbb{R}_+)} + F_i F_j \mathrm{d}_\pi^*(\nabla_{h_i} h_j) + F_i \mathrm{d}_\pi^*(h_j)\mathrm{d}_{h_i}F_j - F_i \mathrm{d}_{h_i}\mathrm{d}_{h_j}F_j)d\pi(\gamma) \\
&= \int_\Gamma (F_i F_j \langle h_i, h_j\rangle_{L^2(\mathbb{R}_+)} + \mathrm{d}_{\nabla_{h_i}h_j}(F_i F_j) + \mathrm{d}_{h_j}(F_i \mathrm{d}_{h_i}F_j) - F_i \mathrm{d}_{h_i}\mathrm{d}_{h_j}F_j)d\pi(\gamma) \\
&= \int_\Gamma (F_i F_j \langle h_i, h_j\rangle_{L^2(\mathbb{R}_+)} + \mathrm{d}_{\nabla_{h_i}h_j}(F_i F_j) + \mathrm{d}_{h_j}F_i \mathrm{d}_{h_i}F_j + F_i(\mathrm{d}_{h_j}\mathrm{d}_{h_i}F_j - \mathrm{d}_{h_i}\mathrm{d}_{h_j}F_j))d\pi(\gamma) \\
&= \int_\Gamma (F_i F_j \langle h_i, h_j\rangle_{L^2(\mathbb{R}_+)} + \mathrm{d}_{\nabla_{h_i}h_j}(F_i F_j) + \mathrm{d}_{h_j}F_i \mathrm{d}_{h_i}F_j + F_i \mathrm{d}_{\nabla_{h_j}h_i - \nabla_{h_i}h_j}F_j)d\pi(\gamma) \\
&= \int_\Gamma (F_i F_j \langle h_i, h_j\rangle_{L^2(\mathbb{R}_+)} + F_j \mathrm{d}_{\nabla_{h_i}h_j}F_i + F_i \mathrm{d}_{\nabla_{h_j}h_i}F_j + \mathrm{d}_{h_j}F_i \mathrm{d}_{h_i}F_j)d\pi(\gamma) \\
&= \int_\Gamma \left(F_i F_j \langle h_i, h_j\rangle_{L^2(\mathbb{R}_+)} + F_j \int_0^\infty \mathrm{d}_s F_i \int_0^\infty \nabla_t h_j(s) h_i(t) dt ds \right. \\
&\quad \left. + F_i \int_0^\infty \mathrm{d}_t F_j \int_0^\infty \nabla_s h_i(t) h_j(s) ds dt + \int_0^\infty h_i(t)\mathrm{d}_t F_j \int_0^\infty h_j(s)\mathrm{d}_s F_i ds dt \right) d\pi(\gamma),
\end{aligned}$$

where we used the commutation relation satisfied by the gradient d:

$$\mathrm{d}_u \mathrm{d}_\pi^* v = \mathrm{d}_\pi^* \nabla_u v + \langle u, v\rangle_{L^2(\mathbb{R}_+)}, \quad u, v \in \mathcal{C}_c^\infty(\mathbb{R}_+), \tag{3.11}$$

which can be proved as follows:

$$\begin{aligned}
\mathrm{d}_u \mathrm{d}_\pi^* v &= -\sum_{k=1}^\infty \dot{v}(T_k) \int_0^{T_k} u(s) ds = -\mathrm{d}_\pi^* \left(v(\cdot) \int_0^\cdot u(s) ds \right) - \int_0^\infty \dot{v}(t) \int_0^t u(s) ds dt \\
&= \mathrm{d}_\pi^*(\nabla_u v) + \langle u, v\rangle_{L^2(\mathbb{R}_+)}.
\end{aligned}$$

□

Finally we state a Weitzenböck type identity on configuration space, that can be read as

$$\mathrm{d}\mathrm{d}_\pi^* + \mathrm{d}^*\mathrm{d} = \nabla^*\nabla + \mathrm{Id}_{L^2(\mathbb{R}_+)},$$

i.e. the Ricci tensor under the Poisson measure is the identity $\mathrm{Id}_{L^2(\mathbb{R}_+)}$ on $L^2(\mathbb{R}_+)$.

Theorem 3.1 *We have for $u \in \mathcal{U}$:*

$$\int_\Gamma (\mathrm{d}_\pi^* u)^2 d\pi(\gamma) + \int_\Gamma \|\mathrm{d}u\|^2_{L^2(\mathbb{R}_+)\wedge L^2(\mathbb{R}_+)} d\pi(\gamma) \tag{3.12}$$
$$= \int_\Gamma \|u\|^2_{L^2(\mathbb{R}_+)} d\pi(\gamma) + \int_\Gamma \|\nabla u\|^2_{L^2(\mathbb{R}_+)\otimes L^2(\mathbb{R}_+)} d\pi(\gamma).$$

Proof. Relation (3.12) for $u = \sum_{i=1}^n h_i F_i \in \mathcal{U}$ follows from (3.9) and Lemma 3.2.

□

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